

Graphs in R

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1 Using `ggm`, `grBase`, `igraph` together

Using **gRbase**

Adjacency matrix $X \perp\!\!\!\perp Y$

```
A <- ug(~a*b, ~b*c*d, ~e)
as(A, "matrix")
```

```
  a b c d e
a 0 1 0 0 0
b 1 0 1 1 0
c 0 1 0 1 0
d 0 1 1 0 0
e 0 0 0 0 0
```

Using **ggm**

Adjacency matrix and plot

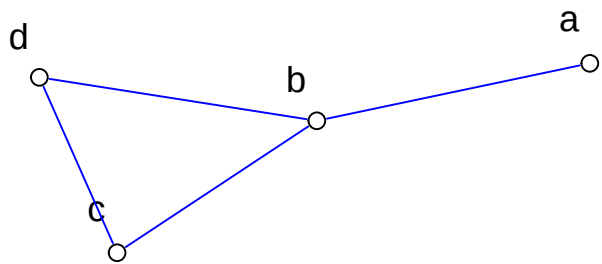
```
G <- UG(~a*b + b*c*d + e)
G
```

```
  a b c d e
a 0 1 0 0 0
b 1 0 1 1 0
c 0 1 0 1 0
```

```
d 0 1 1 0 0
e 0 0 0 0 0
```

```
drawGraph(G, layout = layout_nicely)
```

e
○



2 Find cliques

With `**gRbase*`

```
get_cliques(A)
```

```
[[1]]
[1] "e"
```

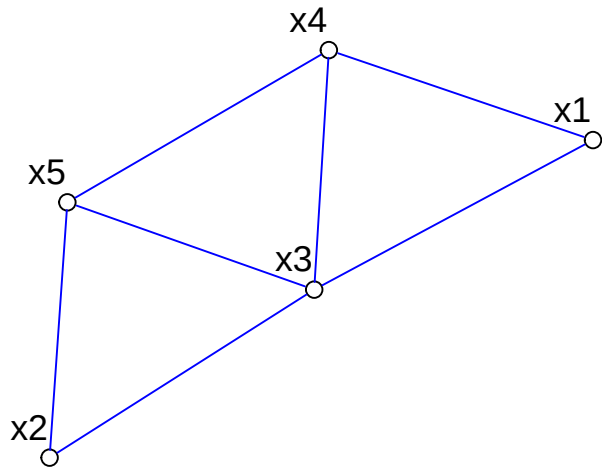
```
[[2]]
[1] "a" "b"
```

```
[[3]]
[1] "b" "c" "d"
```

3 Separation

With **ggm** and **igraph**

```
G <- UG( ~ x1*x3*x4 + x2*x3*x5 + x3*x4*x5)
drawGraph(G, layout = layout_nicely)
```



With **gRbase** and **igraph**

```
g <- graph_from_adjacency_matrix(G, mode = "undirected")
separates("x5", "x1", "x4", g)
```

```
[1] FALSE
```

```
separates("x5", "x1", c("x4", "x3"), g)
```

```
[1] TRUE
```

```
get_cliques(g)
```

```
[[1]]
```

```
[1] "x1" "x3" "x4"
```

```
[[2]]
```

```
[1] "x3" "x5" "x4"
```

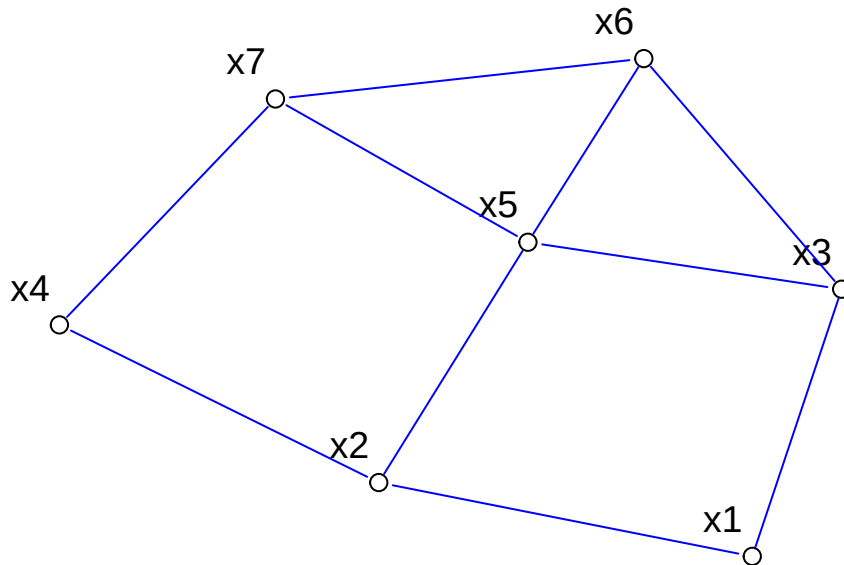
```
[[3]]
```

```
[1] "x3" "x5" "x2"
```

4 Exercise

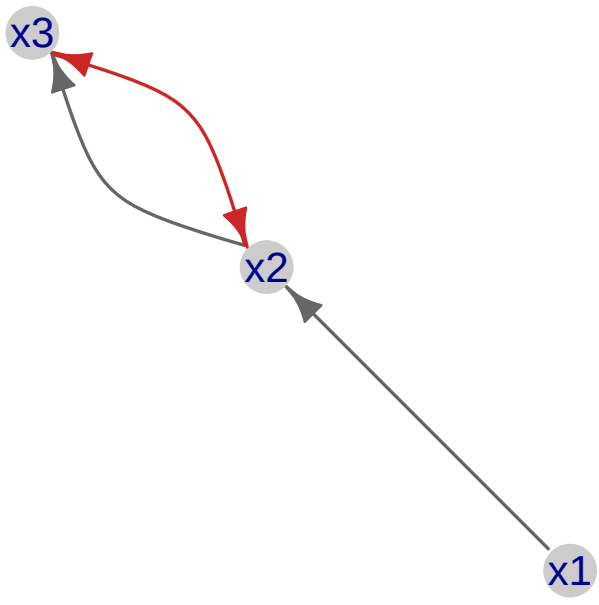
Lauritzen's graph

```
G <- UG(~ x1*x2 + x2*x5 + x5*x3 +  
        + x1*x3 + x2*x4 + x4*x7 +  
        + x7*x5 + x3*x6 + x5*x6 + x7*x6)  
drawGraph(G, layout = layout_nicely)
```



5 Plot with ggm

```
source("~/Documents/R_packages/ggm/plotGraph2.R")  
D <- makeMG(dg = DAG(x2 ~ x1 + xH, x3 ~ x2 + xH))  
options(repr.plot.width = 4, repr.plot.height = 4)  
sg <- SG(D, M = "xH")  
plotGraph2(sg, tcltk = FALSE, dashed = FALSE)
```



<i>Age</i>	<i>Frequency</i>
18 – 25	15
26 – 35	33
36 – 45	22